



ENTERPRISE SYSTEMS
PRODUCT · OPERATIONS · ENGINEERING

● CODE-VERIFIED PRODUCT BLUEPRINT

CRM Enterprise Architecture & Intelligence

A complete product, business-logic and technical blueprint—from customer acquisition to compliance delivery, connected securely with CCP.

PRODUCT MAP

DATA & APIs

WORKFLOWS

CCP INTEGRATION

ROADMAP

System

CRM V2 · Product & Technology Report

Evidence base

Frontend, backend, models, services, tests and Git history

Edition

13 July 2026

Classification

Internal · Confidential

Executive Summary

This CRM is a full-stack operational platform that joins lead acquisition, client onboarding, quotation, approval, compliance/annual-return processing, user/team administration, notifications and scheduled work. It uses a React single-page application, an Express service layer, MongoDB persistence and a protected integration boundary with CCP.

11**Application routes**

Including authentication, dashboards and business workspaces

10**Core data models**

User, Team, Lead, Client, Quote, Return, Approval, Notification, Calendar, Role

7**Business roles**

Operation, Admin, Super Admin, Manager, Compliance, Sales, Accounts

Product value: one operating surface provides visibility from first lead to client compliance delivery, while approval gates and reporting scopes keep ownership controlled.

How to read this report

Product & navigation	What each screen contains and why it exists
Business logic	Status transitions, ownership rules, approvals and reports
Architecture	Frontend → API → database → CCP data movement
Data dictionary	Meaning of major entities and fields
Update register	Verified Git changes plus the profile-menu fix in this working tree
Recommendations	Enterprise-hardening backlog, explicitly separated from current behavior

Accuracy note: “Current implementation” statements are derived from repository code. “Recommended” statements describe a future enterprise target and are not claimed as already deployed.

Navigation & Information Architecture

Home / Operations

Dashboard · Pending Approval · Notifications · Calendar · User Management

Sales

Lead Generation · Client Master · Add Quotation

Deep Workspaces

Client Data Processing · Annual Returns · Profile & Security

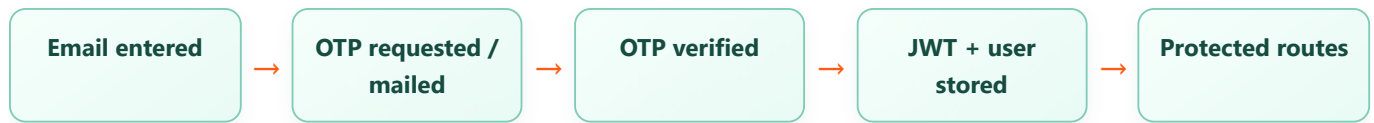
Route register

Route	Page	Meaning	Protection
/	Login	Email-based OTP request entry	Public
/verify	Verify OTP	OTP verification, token and user session creation	Public
/dashboard	Dashboard	Role-aware operational/sales intelligence	JWT
/dashboard/users	User Management	People, roles, teams and reporting hierarchy	JWT + admin actions gated by API
/pending-approval	Approval Desk	Client and quotation approval queue	JWT; mutation admin-gated
/notifications	Notifications	Publish, filter, pin and download notices	JWT; editing privileged
/calendar	Calendar	Todos, follow-ups, assignment and history	JWT
/sales/lead-generation	Leads	Lead capture, import/export and follow-up	JWT
/sales/client-master	Clients	Client directory and onboarding records	JWT
/sales/quotations	Quotation	Create/update commercial proposals	JWT
/sales/client-data-processing/:clientKey/:annualYear	Annual workspace	Year-specific compliance data processing	JWT

Shared chrome

Sidebar groups work by Operations and Sales. **Topbar** provides information, notification indicator, identity, Profile Settings and Logout. **DashboardShell** coordinates responsive sidebar state, collapsed desktop mode and session logout.

Authentication & Session



Login page

- Accepts the user's registered email.
- Calls `POST /auth/request-otp`.
- Moves the user to verification after successful request.
- Designed to avoid sharing passwords during normal login.

Verify OTP page

- Verifies the time-bound code through `/auth/verify-otp`.
- Supports OTP resend.
- Stores authenticated user and token for API calls.
- `ProtectedRoute` blocks unauthenticated business pages.

Security meaning

Term	Explanation
JWT	A signed session token sent as <code>Authorization: Bearer ...</code> . Backend verifies it and loads the active user.
OTP expiry	A verification code is useful only within a defined time window, reducing replay risk.
Active user check	Even a structurally valid token is rejected when its user is missing or inactive.
Role guard	<code>requireRoles</code> returns HTTP 403 when the authenticated role cannot perform an action.

Enterprise recommendation: production must set a strong `JWT_SECRET`; remove reliance on the development fallback, add refresh/session revocation, audit login events and enforce OTP throttling.

Dashboard & Reports

The dashboard is role-aware: it converts raw leads, clients, quotations, approvals, annual returns and calendar follow-ups into management views. Operations and Sales users see different KPIs and task-oriented panels.

Operations dashboard

- Operational workload and team-level visibility.
- Client/application and annual-return progress.
- Pending actions and follow-up timeline.
- Approval status and annual processing indicators.
- Calendar drill-through for scheduled work.

Sales dashboard

- Lead pipeline, today leads and conversion measures.
- Quotation and client conversion views.
- Assigned-user and manager hierarchy scoping.
- Follow-ups and overdue attention.
- Interactive report modals with record rows.

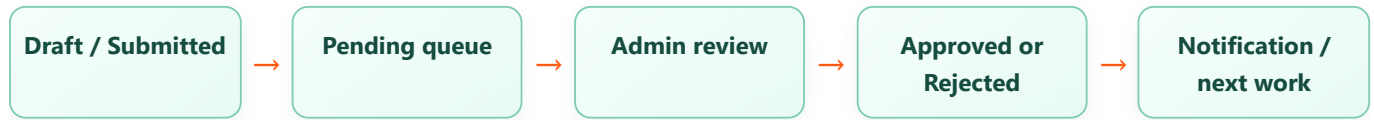
Report logic

Report/KPI	Meaning	Primary source	Expected action
Lead volume	Total visible lead records	CRM + normalized CCP leads	Review capacity and ownership
Today leads	Leads created/dated today	Lead timestamps/date	Immediate qualification
Conversion	Lead/client progression indicator	Lead and client sources	Improve pipeline movement
Pending approvals	Items waiting for authorized decision	Clients and quotations	Approve/reject with remarks
Annual progress	Year-wise compliance processing state	AnnualReturn	Complete missing sections/review
Follow-ups	Scheduled customer touchpoints	Calendar + lead history	Open Calendar and complete

Visibility logic: server utilities build a user scope from role, direct ownership, manager and operation-head relationships. This prevents every user from automatically receiving an unrestricted organizational dataset.

Pending Approval

The Approval Desk is a control gate between record preparation and operational acceptance. It combines client and quotation queues and allows authorized admins to decide individually or in bulk.



Screen components

Component	Purpose
Summary metrics	Fast count of waiting client and quotation decisions.
Filters/tabs	Separate work type and status so reviewers focus on the right queue.
Detail review	Shows submitted record context before the decision.
Approve/reject	Updates approval status; API checks the user's admin role.
Approve all	Bulk-completes pending approvals through dedicated PATCH endpoints.
Reminder service	Backend scheduler supports pending approval reminders after database startup.

Governance meaning: maker-checker separation reduces accidental publication. A creator prepares data; an authorized reviewer controls acceptance.

Notifications

What the page contains

- Search, tag and Active/Inactive filters.
- Comfortable/compact density.
- Multi-select and bulk actions.
- Create/edit/view notification modal.
- Pin/unpin priority notices.
- Attachment display and download.

Business meaning

- **Tag:** subject classification such as Training, SOP, Company Profile or Policy.
- **Status:** whether a notice is currently usable/visible.
- **Pinned:** lifts important content above chronological order.
- **Attachment:** evidence or reference material linked to the notice.

Permissions and CCP

Authenticated users can list notices. Create/update is restricted to Admin, Super Admin and Manager. Dedicated secret-protected endpoints allow CCP to list or synchronize notices, while outbound notification sync uses a configured CCP endpoint.

Update completed in this working tree: Notifications now passes `onOpenProfile` to the shared shell, mounts ProfileModal, supports profile/password updates and maintains user state in local storage. The previously unresponsive Profile Settings click is fixed.

Calendar & Todo Management

Calendar joins customer follow-ups with internal tasks. It offers month/week/day contexts, agenda/timeline views, assignment, revision, completion remarks and history.

Todo

Internal action with title, description, priority, category, scheduled date/time and assignee.

Follow-up

Customer-facing touchpoint connected to a lead or client.

History

Change trail recording reassignment, revision or completion context.

Logic

Rule	Result
Status = completed	Item receives completed tone and 100% progress.
Scheduled date before today	Open item becomes overdue and demands attention.
History exists	Item can be represented as revised.
Assignment	User and reason are captured to support accountability.
Storage	Server CRUD is backed by CalendarItem; local storage is used as resilience/cache behavior.

Client and lead selectors merge CRM records with CCP results, so scheduled work can reference the shared operational universe.

Update completed in this working tree: Calendar now opens Profile Settings correctly and includes working profile save, password update and logout behavior.

User & Team Management

User Management controls identity, access level, team placement and reporting hierarchy. Admins can create and edit users, inspect details, activate/deactivate accounts and create teams.

Role	Meaning	Typical scope
Operation	Executes operational work	Assigned clients, tasks and compliance work
Admin	System administrator	Users, teams, approvals and broad visibility
Super Admin	Highest configured administrative role	Organization-wide control
Manager	Supervisory role	Team/linked user visibility and notices
Compliance	Compliance manager/reviewer	Annual and regulatory workflow
Sales	Commercial pipeline owner	Leads, clients, quotes, follow-ups
Accounts	Financial participant	Quotation/financially relevant views

Hierarchy fields

`teamId` assigns organizational unit; `managerId` defines direct reporting; `operationHeadId` provides operational leadership scope; `createdBy` records provenance. Active state immediately affects authentication.

CCP user/team synchronization

CRM can push create/update user payloads and team payloads to CCP. The returned `ccpUserId` is saved when synchronization succeeds. Shared-secret endpoints also permit controlled inbound synchronization.

Lead Generation

This page captures prospects, qualifies them and schedules the next commercial action. It combines manually created CRM leads, Excel imports and normalized CCP leads.

Core sections

- Lead listing, search, filters and pagination.
- Create/edit form with company, category, service, address and contact data.
- Draft versus submitted workflow.
- Assigned owner and referral/source data.
- Follow-up date, time, remarks and history.

Data movement

- Excel `.xlsx/.xls` headers are mapped into lead fields.
- Bulk endpoint persists imported rows.
- Export creates a workbook for reporting.
- CCP rows are normalized to one common lead shape.
- Duplicate/identity handling uses lead code and source identifiers.

Lead lifecycle



Reporting: exported sheets, status filters, assigned-owner analysis, dashboard pipeline counts and follow-up timelines together form the current reporting surface.

Client Master

Client Master is the authoritative operational directory after a prospect becomes a client. It supports detailed onboarding, administrative controls, approval, import and annual compliance work.

Major information groups

Group	Examples and meaning
Basic identity	Legal/trade name, client unique ID, categories and industry.
Registered address	Address, city, state and PIN for formal correspondence.
Contact	Named contact, designation, email and mobile details.
Regulatory	GST, CPCB registration, EPR/PIBO categories and related compliance data.
Admin controls	Approval status, visibility status and assigned user.
Workflow	Draft allows continued preparation; submitted enters controlled processing.

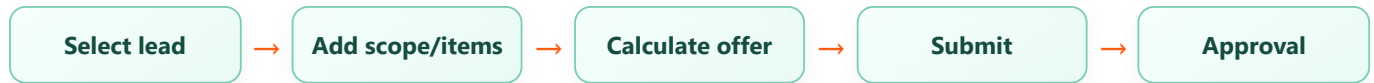
The directory merges CRM and CCP clients, while utility functions normalize nested client data and stable unique identifiers. Excel bulk import supports high-volume onboarding.

Client-level interactions

Follow-ups and todos share Calendar data. Annual-return history opens year-specific processing. Client approval is controlled through the Approval Desk.

Quotation & Commercial Workflow

Quotation turns a qualified lead into a structured commercial offer. It retains lead context, line items, pricing and approval state.



Object	Meaning
Lead details	Commercial recipient and source context.
Quote items	Individual services/products and financial values.
Status/workflow	Draft, submission and approval progression.
Approval remarks	Reviewer rationale for decision.

The page can consume CCP lead options. Backend exposes list, create, update, individual approval and approve-all-pending endpoints.

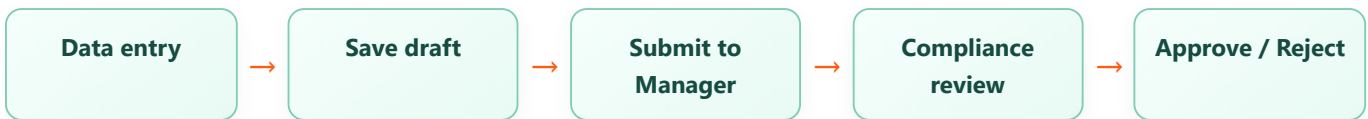
Annual Returns & Compliance Workspace

Annual Returns provides year-wise regulatory processing for each client. The workspace remembers active tab/section, filing data, approval workflow and saved time.

Workspace structure

Tab/part	Purpose
Basic Info	Client identity and year context required for the filing.
Financials	Quotation, SLA, receipt and finance-related dates/values.
Data	Structured regulatory inputs and evidence.
Part A — Client Data	Core applicant and registration context.
Part B — Consent Details	Water/air consent validity and related records.
Part C — Raw Data & Interaction	Operational source data and interactions.
Part D — Plant & Declaration	Facility and formal declaration inputs.
CPCB Letter	Authority-specific letter preparation and dates.

Review sequence



Persistence: flexible nested MongoDB fields support evolving compliance schemas while indexed client/year/status fields support lookup. This flexibility should be paired with API-level validation as the product matures.

Technical Architecture

Experience layer · React 18 · React Router · Tailwind/CSS · Framer Motion · GSAP · Recharts · Lucide



Client service layer · Axios · centralized API endpoints · CCP normalization · local cache/fallback

↓ **HTTPS / JSON**

Application layer · Node.js · Express · route/controller separation · auth & role middleware · schedulers

↓ **Mongoose ODM**

Data layer · MongoDB Atlas/local MongoDB · indexed entity collections · timestamps

↔ **Shared secret / server-to-server JSON**

External platform · CCP leads · clients · users · teams · notifications

Why this structure works

- **SPA:** fast navigation and reusable shared shell without full reloads.
- **REST API:** clear resource boundaries and independent frontend/backend deployment.
- **Mongoose:** schema, references, indexes and timestamps over MongoDB.
- **Normalization:** CCP payload variants are converted to consistent frontend fields.
- **Serverless readiness:** Vercel entry points coexist with normal Express startup.

Deployment view

Browser



Vercel Frontend



Express API

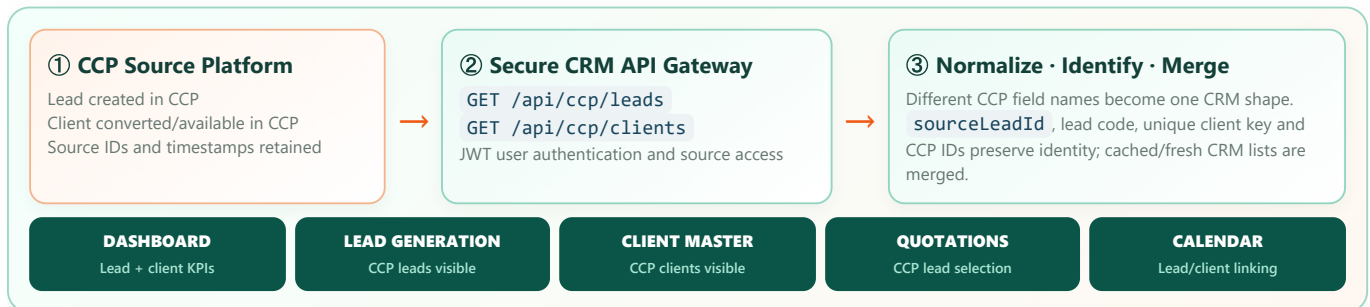


MongoDB

How CRM Is Linked to CCP

Implemented business flow: a Lead created in CCP is read into CRM and appears in Lead Generation and Dashboard. A Client available in CCP is read into CRM and appears in Client Master and Dashboard. CRM does not blindly duplicate the screen data; it normalizes CCP fields and merges them with CRM-owned records for one unified view.

Inbound master-data pipeline: CCP → CRM



Exactly what happens to CCP records

CCP event/data	CRM handling	Where the user sees it
New CCP lead	Backend exposes it through the authenticated CCP lead endpoint; frontend normalizes company, status, categories, address, contact, source and assignment fields.	Lead Generation list, Dashboard lead reports/KPIs, Quotation lead selector and Calendar lead selector.
CCP client	Backend returns the CCP client collection; client utilities derive a stable unique ID, read nested client data and merge it with CRM client records.	Client Master directory, Dashboard client analytics and Calendar client selector.
Same identity in CRM and CCP	Merge/identity utilities reduce duplicate display by using source IDs, CRM IDs, CCP IDs, lead codes and client unique keys.	Unified page dataset rather than separate disconnected tables.
CCP temporarily unavailable	Pages use cached/fresh-list helpers or local CRM data where implemented and surface source/failure state.	CRM remains usable with the latest available data, subject to page behavior.

Outbound synchronization: CRM → CCP

Entity	Default sync target	Trigger
User	/api/crm/users/sync	Admin create/update and explicit sync
Team	/api/crm/teams/sync	Team create/update synchronization
Notification	/api/crm/notifications/sync	Notification create/update synchronization

Trust boundary

`CCP_SHARED_SECRET` is transmitted as `x-ccp-secret` for protected machine-to-machine calls. Human browser calls still require JWT. Sync helpers return structured success/failure results so CRM actions can complete while reporting CCP sync status.

Enterprise recommendation: add an outbox queue with retry/backoff, idempotency keys, correlation IDs, signed requests, sync dashboards and dead-letter handling. Direct synchronous HTTP alone is vulnerable to temporary CCP downtime.

Core Data Model

Entity	Important fields	Relationship / purpose
User	email, CRM/CCP IDs, role, team, manager, operation head, active	Identity, access and reporting hierarchy
Team	name, description, active	Organizational grouping
Lead	leadCode, sourceLeadId, company, status, contacts, assignment, follow-up	Prospect pipeline
Client	selectedLead, adminControls, data, workflowStatus	Approved/operational customer record
Quotation	lead details, items, status, approval	Commercial offer
AnnualReturn	clientKey, annualYear, sections, filings, approvalWorkflow	Year-specific compliance processing
PendingApproval	entity/type/status/reminders	Approval control and reminder state
Notification	title, description, tag, status, attachment, pinned	Organizational communication
CalendarItem	type, schedule, assignment, status, history	Todo and follow-up execution

MongoDB design meaning: references model stable relationships, while Mixed objects accommodate evolving compliance forms. High-value identifiers such as email, leadCode and clientKey are unique/indexed.

API Inventory & Responsibilities

Resource	Methods	Responsibility
Auth	POST OTP/resend/verify; GET/PUT me; admin user CRUD/sync	Identity, session and user administration
Leads	GET, POST, POST bulk, PUT :id	Pipeline records and imports
Clients	GET, POST, bulk, PUT, PATCH approval, annual return	Onboarding, approval and client data
Quotations	GET, POST, PUT, PATCH approval/bulk approval	Commercial workflow
Annual returns	GET	Cross-client year/status listing
Notifications	GET, POST, PUT; CCP list/sync	Communication content and exchange
Calendar	GET, POST, PUT, DELETE	Persistent todos and follow-ups
Teams	GET, POST; CCP list/sync	Organization structure
CCP proxy	GET leads/clients	Authenticated external-data access

Error semantics

401 means token/secret missing or invalid. **403** means identity is valid but lacks role permission. **503** reports database startup/unavailability. Controller validation failures use client-error responses; unexpected failures use server-error responses.

Reporting Catalogue

Area	Current report/view	Decision supported	Export
Dashboard	Role KPIs, pipeline, operations, follow-up rows	Where management attention is needed	On-screen drill-down
Leads	Filtered list, status/ownership, follow-up history	Qualification and conversion management	XLSX
Clients	Directory, approval and annual history	Onboarding and delivery readiness	XLSX import; directory reporting
Approvals	Pending client/quote queues and totals	Maker-checker turnaround	Operational view
Calendar	Today, upcoming, overdue, completed, timeline	Execution and SLA discipline	On-screen
Annual Return	Year/status/client workspace progress	Compliance completion and review	Structured saved record
Notifications	Active/inactive, tag and pinned views	Communication governance	Attachment download

Recommended enterprise report pack

- Lead funnel by source, owner, category and aging.
- Approval turnaround time and rejection reason analysis.
- Quotation win rate, value, aging and approval SLA.
- Client onboarding cycle time and data completeness.
- Annual-return completion by FY, category, SPOC and risk.
- Overdue follow-up heatmap and team completion rate.
- CCP synchronization success, retry count and data reconciliation.



Verified Update Register

Date	Commit	Verified change theme
11 Jul 2026	f3d0f47	CRM dashboards, annual workspace and attachments improved
10 Jul 2026	50bce18	Accounts role added; calendar items persisted
08 Jul 2026	8fd4a53	Stale SPA asset caching prevented on Vercel
08 Jul 2026	ea91b9f	Production verify-page chunk loading fixed
08 Jul 2026	0e39ad1	Dashboard UI and approval flows polished
02 Jul 2026	223e70d	Dashboard donut charts animated
02 Jul 2026	25e889a	Scroll reset on route changes
02 Jul 2026	aeфа869 and related	Annual review drawer content, overlay, sizing and scroll behavior improved
13 Jul 2026	Working tree	Notifications and Calendar Profile Settings now open and provide save/password/logout flow

Change-control note: the last row is an uncommitted working-tree update at report generation time; it is verified by a successful Vite production build.

Testing, Reliability & Observability

Current quality evidence

- Backend Node test suites cover client controller logic, calendar item controller behavior and Accounts role behavior.
- Frontend production build successfully transforms the complete application.
- Database readiness middleware returns a controlled 503 instead of silently serving broken API requests.
- Uncaught exception and unhandled rejection logging exists at process level.
- Local cache/fallback behavior protects selected pages from short-lived data-source failures.

Recommended quality gates

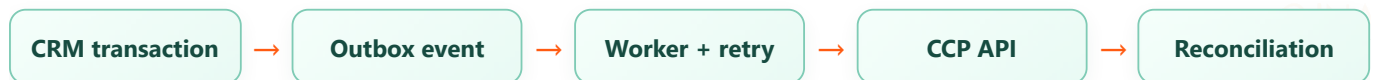
Gate	Target
Unit tests	Controllers, normalization, permission scope and calculations
API integration tests	Auth, roles, CRUD, approval transitions, CCP secrets
UI tests	Login, profile, lead-to-client, quotation approval, annual review
Contract tests	CRM ↔ CCP payloads and version compatibility
Performance tests	Large lead/client lists, Excel import and dashboard aggregation
Monitoring	Structured logs, error tracking, health checks, latency and sync metrics

Big-Company Architecture Recommendations

• Current foundation • Recommended target

Priority	Recommendation	Business outcome
P0	Move all fallback secrets out of code; rotate JWT/CCP credentials; validate production environment at boot	Reduces critical security exposure
P0	Introduce immutable audit events for login, user changes, approvals and annual-return decisions	Accountability and compliance evidence
P1	CCP outbox, retry/backoff, idempotency and reconciliation dashboard	Reliable cross-system consistency
P1	Server-side pagination, filtering and indexed dashboard aggregation	Scales beyond browser-sized datasets
P1	Schema validation/versioning for nested Client and AnnualReturn objects	Data quality without losing form flexibility
P1	Object storage for attachments with signed URLs, virus scan and retention rules	Secure document handling
P2	Formal RBAC permission matrix instead of role-name checks alone	Fine-grained, auditable authorization
P2	BI warehouse/read model for executive reports	Fast analytics without transactional load
P2	CI/CD gates: lint, test, build, dependency scan and migration checks	Safer frequent releases
P3	Accessibility, localization and design-system component governance	Consistent enterprise UX

Target integration pattern



One-by-One Meaning of Key Terms

Term	Plain-language explanation
CRM	System of record for customer acquisition and service operations.
CCP	Connected external platform exchanging leads, clients, users, teams and notifications with CRM.
SPA	Single Page Application; navigation updates the view without reloading the whole website.
API	Defined interface through which frontend or external systems read/change data.
REST	Resource-oriented HTTP design using GET, POST, PUT, PATCH and DELETE.
JWT	Signed authentication token proving the browser session's identity.
RBAC	Role-Based Access Control: permissions depend on the user's assigned role.
OTP	One-Time Password used for time-bound login verification.
PIBO	Producer, Importer and Brand Owner classification used in EPR compliance work.
EPR	Extended Producer Responsibility regulatory/compliance framework.
CPCB	Central Pollution Control Board; authority context for registration and letters.
SPOC	Single Point of Contact responsible for coordination.
Maker-checker	One person prepares; another authorized person approves.
Normalization	Converting different source field names/shapes into one consistent structure.
Idempotency	Repeating the same integration request does not create duplicate business effects.
Outbox	Reliable table/collection of integration events waiting to be delivered.
Index	Database structure that accelerates lookup and may enforce uniqueness.
Mixed field	Flexible MongoDB object used where form structure can evolve.

Configuration Checklist

Configuration	Purpose	Production expectation
MONGO_URI / DB_NAME	MongoDB connection and database	Private managed secret and restricted network
JWT_SECRET	Signs/verifies sessions	Long rotated secret; never fallback
CLIENT_ORIGIN	CORS allowed origins	Exact production domains
CCP_SHARED_SECRET	Machine trust between CRM and CCP	Rotated vault-managed secret
CCP sync URLs	User/team/notification delivery	Environment-specific HTTPS endpoints
Mail settings	OTP and reminder email	Authenticated provider, SPF/DKIM/DMARC

Operational runbook

1. Verify database connectivity and API root health.
2. Verify OTP mail provider and delivery.
3. Test one admin and one limited-role login.
4. Test CCP lead/client read and one safe sync.
5. Check pending approval scheduler startup logs.
6. Build frontend and confirm current asset deployment.
7. Review error logs and synchronization failures.

Traceability & Sign-off

This report was produced by inspecting the application routes, navigation constants, page components, service endpoints, CCP adapters, Express application, backend routes/controllers, Mongoose models, authentication middleware, synchronization utilities, tests, package manifests and Git history.

Build verification: `npm.cmd run build` completed successfully with Vite 5.4.21 and 2,667 transformed modules after the Profile Settings fix.

Ownership matrix template

Area	Business owner	Technical owner	Review frequency
Sales pipeline	Sales Head	CRM Engineering	Monthly
Client & annual compliance	Operations/Compliance	CRM Engineering	Monthly
Approvals	Admin/Management	CRM Engineering	Quarterly
CCP integration	Product Operations	Integration Owner	Monthly
Security & access	System Owner	Security/Engineering	Quarterly

Prepared / reviewed by

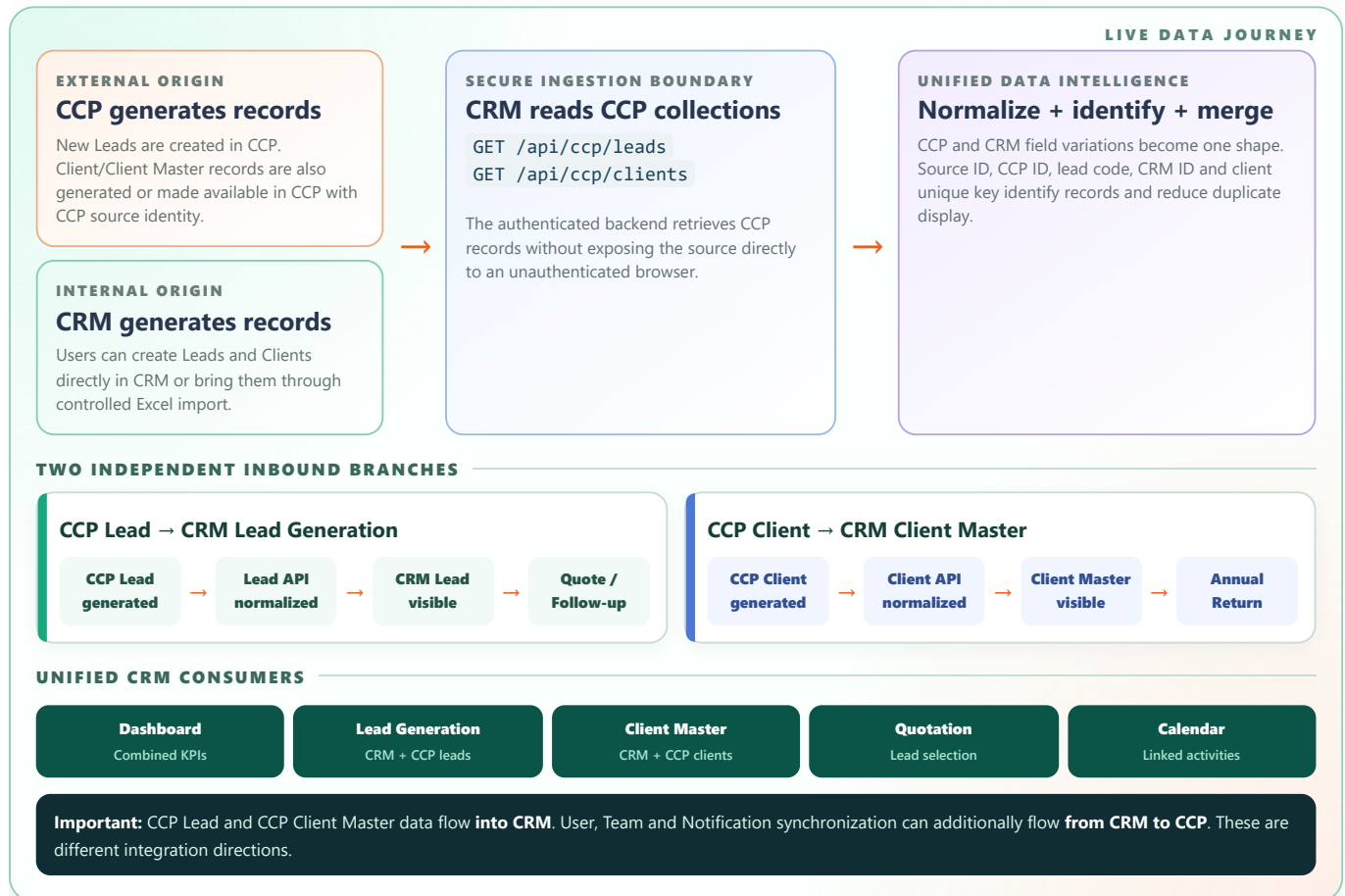
Approved by

End of report · CRM V2 Enterprise Product, Logic & Architecture Report · 13 July 2026



CCP + CRM Lead & Client Master Journey

This roadmap separates **where a record is generated** from **where it is consumed**. Both CCP and CRM can originate business data; CRM presents one unified operational view.



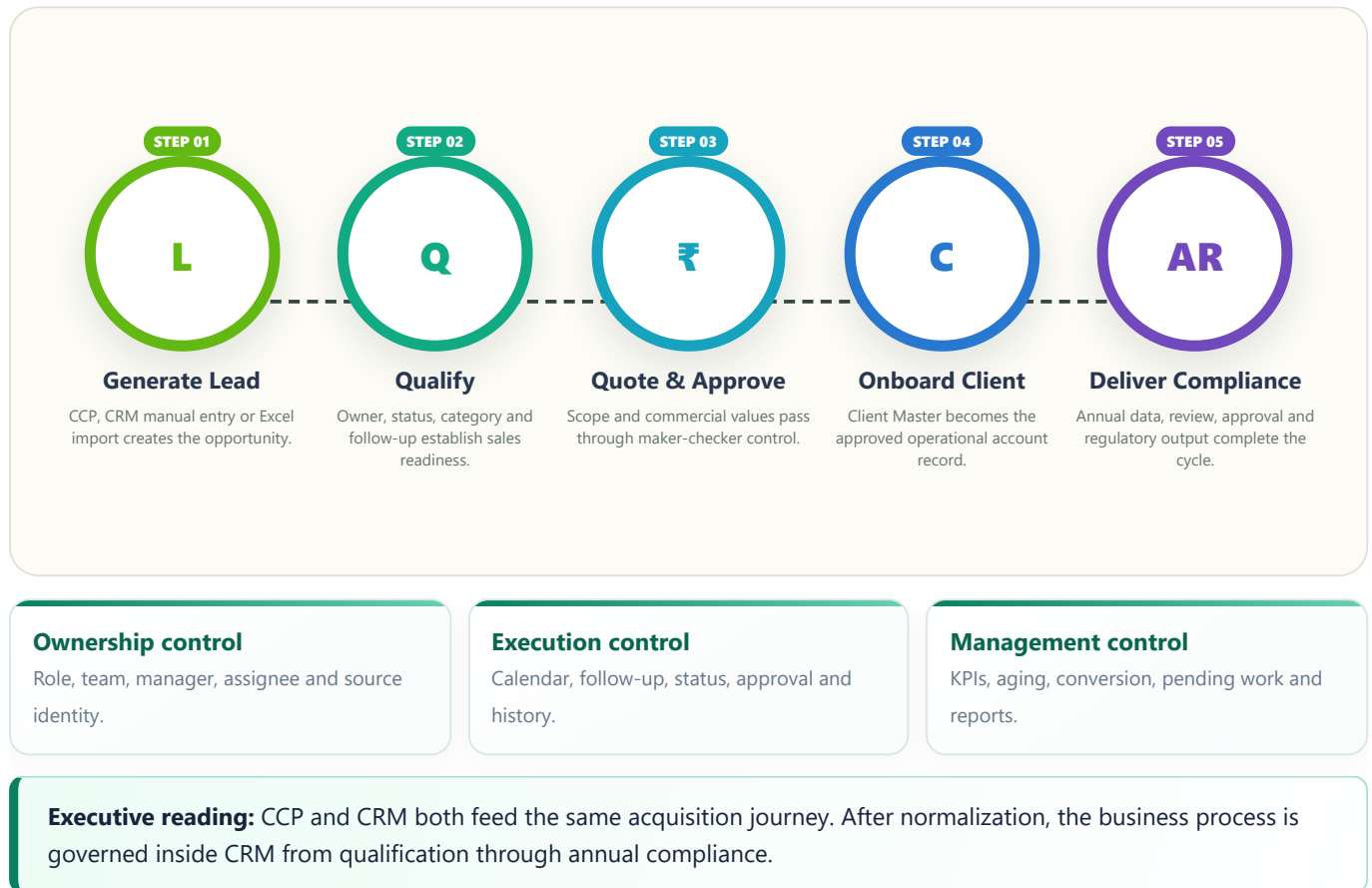
End-to-end business roadmap



Technical meaning of “comes into CRM”: CCP collections are securely read, normalized and merged into CRM page datasets. This does not automatically mean every external row is copied into the CRM MongoDB collection; persistence happens only through the relevant create/import workflow.

Customer-to-Compliance Process Flow

A single high-quality executive roadmap replaces the four decorative infographic screens. It shows the real business sequence and the control attached to every stage.



Relationship map

Old enterprise topology UI retained with the approved whiteboard workflow: CCP is the external source, CRM governs people and records, and both commercial and compliance work converge at one approval control.

